

ERIC BOLTON

New York, NY · edbltn@gmail.com · www.ericdidierbolton.com · github.com/edbltn · linkedin.com/in/edbltn

EXPERIENCE

Spherical Harmonics

Dec 2025–Present

Co-founder

- Built City Edit, the nonprofit's open-source crowdsourced map for voting on safer streets, live in 9 cities incl. NYC and D.C.
- Fit a pedestrian-volume model (leave-one-out R^2 0.71) to rank 11,907 NYC signalized crossings by person-hours spent waiting.
- As sole developer, built a custom multi-agent coding harness (Claude Code, Codex) that orchestrates reviewer-gated workers.
- Produced detailed commit logs and debug-mode code walkthroughs per major release to deeply understand agent-written code.

Meta

Apr 2025–May 2026

Software Engineer, Instagram Reels

- Co-led development of a distilled multimodal model for time-sensitive Reels, lifting precision from 0.752 to 0.889 at 80% recall.
- Owned the model's ML lifecycle: training data, architecture, evaluation, feature-store integration, and experiments.
- Wrote a trend-expansion job in Hack and limited retrieval to feature-covered media, freeing capacity without losing recall.
- Created reusable evaluation pipelines with representative sampling, automated LLM scoring, and quality dashboards.

NewsWyze

Feb 2024–Apr 2025

CTO

- Led a cross-functional team integrating LLM summarizers into a scalable, observable production system for newsrooms.
- Built a RAG pipeline on journalistic annotations to improve model outputs, with evaluation and monitoring in production.
- Delivered culturally aware LLM translations, cutting editing time by 85% across target languages.

Applied XL

Jun 2021–Feb 2024

Founding Engineer

- Managed a two-person ML team building data-to-text LLMs for biotech investors, fine-tuned with LoRA and DPO.
- Built an LLM evaluation framework using BERTScore, FActScore, and custom scorer prompts to track output quality.
- Served as interim CTO, overseeing the Django API, React frontend, data workflows, and AWS infrastructure.
- Developed deterministic graph clustering and production classifiers for drug-pipeline events.
- Co-authored a 70,000-trial Weibull report finding trials with multiple delays were twice as likely to stop early.

Columbia University, Brown Institute

Jun 2020–Jun 2021

ML Engineer, Local News Lab

- Built a news recommender (NMF, Doc2Vec) balancing recency, relevance, and diversity, adopted by Washington City Paper.
- Prototyped ML-driven paywalls to support alternative revenue models for local newsrooms.
- Ran A/B tests and contextual-bandit experiments to compare recommendation strategies.

The Wall Street Journal

Jun 2018–Jun 2020

Machine Learning Scientist, R&D

- Talk2020: Built an NLP pipeline and search tool for hundreds of thousands of political quotes, published to millions of subscribers.
- AI Portraits: Trained a GAN turning photos into WSJ-style stipple drawings, earning 100,000+ user engagements.
- Story Blueprint: Trained and evaluated topic models and word embeddings on a 1M+ article corpus to inform strategy.
- Conducted research featured in WSJ news articles on topics spanning deep fakes, algorithmic bias, and NLP.
- Data Fellowship: Won Best Intern Project with a data-driven engagement strategy to attract young readers.

SELECTED RESEARCH & PROJECTS

Recurse Center

May 2024–Jul 2024

- Built Frontier Beat, a platform tracking and summarizing trending arXiv papers and associated GitHub repositories.

Columbia University

Sep 2018–May 2019

- Master's Project: Clustered text embeddings to visualize news article bias during Brexit in an interactive interface.
- Columbia Speech Lab: Uncovered factors shaping polarization using matrix factorization on trust ratings in news.
- Structured Debate Generation: Co-authored research reaching near-human debate generation via debate-tree data augmentation.

EDUCATION

Columbia University

May 2019

M.S. Computer Science; M.S. Journalism.

Brown Center Fellow; Machine Learning Teaching Assistant; Honors in Business Essentials and 800 Words.

Princeton University

May 2015

A.B. Geosciences.

TECHNICAL SKILLS

- Languages: Python, SQL, Hack/PHP, Java, C, Go, R
- AI & LLMs: RAG, LoRA/DPO, distillation, LLM-as-a-judge evals, multi-agent orchestration, vLLM, FAISS, MLflow, W&B
- ML & infrastructure: PyTorch, TensorFlow, Transformers, PEFT, TRL, XGBoost, PySpark, AWS, GCP, Terraform, Airflow

LANGUAGES

English and French (native); Spanish (intermediate)